

Statistics

Lecture 26 **Interval Estimations**

Muhammad Mahbub Alam, PhD
Professor
CSE Department, IUT

Confidence Interval

- In frequentist school of thought, a confidence interval is an estimation of an unknown parameter, where
 - The estimation is a Range of values
 - A confidence is associated with the estimation
- It describes the amount of uncertainty associated with a point estimate of a population parameter
- A confidence level indicates the percentage of time true parameter value will lie within the range

Confidence Interval

- To find the confidence interval
 - Need to set the confidence level
 - 90%, 95%, 99%
- A 90% confidence level means
 - Estimated interval include the true parameter value 90% of the time
- In general, we say $(1 - \alpha)\%$ CI
 - α is the significance level

Definition

- Let X be a random sample, with statistical parameter θ
- A $(1 - \alpha)\%$ CI for θ , with confidence level $(1 - \alpha)$
 - is an interval $(L(X), U(X))$
 - With the property
$$P[L(X) < \theta < U(X)] = 1 - \alpha$$

Factors related to CI

- The confidence level
- The sample size
- The variability of the sample
- The distribution of the estimation

confidence interval

$$P(L(X) < \theta < U(X)) = \gamma$$

A larger sample would
produce a
narrower confidence interval.

A higher confidence level would
demand a
wider confidence interval.

Greater variability in the sample
produces a
wider confidence interval,

What exactly is the CI?

A confidence interval is the mean of your estimate
plus and minus
the variation in that estimate.

$$\text{i.e., } CI = \text{Mean} \pm \text{Variation}$$

This is the range of values you expect
your estimate to fall between
if you redo your test,
within a certain level of confidence.

Confidence, in statistics, is another way
to describe probability.

For example, if you construct a confidence interval
with a 95% confidence level,
you are confident that 95 out of 100 times
the estimate will fall between
the upper and lower values
specified by the confidence interval.

How to Calculate?

For calculating the confidence interval,
we need to know

- The point estimate (for which we need to construct the confidence interval)
- The critical values for the test statistic.
- The standard deviation of the sample.
- The sample size.

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \sqrt{n} \frac{(\bar{X} - \mu)}{\sigma}$$

has a standard normal distribution. Therefore,

$$P \left\{ -1.96 < \sqrt{n} \frac{(\bar{X} - \mu)}{\sigma} < 1.96 \right\} = .95$$

or, equivalently,

$$P \left\{ -1.96 \frac{\sigma}{\sqrt{n}} < \bar{X} - \mu < 1.96 \frac{\sigma}{\sqrt{n}} \right\} = .95$$

Multiplying through by -1 yields the equivalent statement

$$P \left\{ -1.96 \frac{\sigma}{\sqrt{n}} < \mu - \bar{X} < 1.96 \frac{\sigma}{\sqrt{n}} \right\} = .95$$

or, equivalently,

$$P \left\{ \bar{X} - 1.96 \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}} \right\} = .95$$

$$\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}}$$

$$\left(\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \right)$$

is called a *95 percent confidence*

Example 7.3.a. Suppose that when a signal having value μ is transmitted from location A the value received at location B is normally distributed with mean μ and variance 4. That is, if μ is sent, then the value received is $\mu + N$ where N , representing noise, is normal with mean 0 and variance 4. To reduce error, suppose the same value is sent 9 times. If the successive values received are 5, 8.5, 12, 15, 7, 9, 7.5, 6.5, 10.5, let us construct a 95 percent confidence interval for μ .

$$\bar{x} = \frac{81}{9} = 9$$

a 95 percent confidence interval for μ is

$$\left(9 - 1.96\frac{\sigma}{3}, 9 + 1.96\frac{\sigma}{3}\right) = (7.69, 10.31)$$

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \sqrt{n} \frac{(\bar{X} - \mu)}{\sigma}$$

has a standard normal distribution. Therefore,

$$P \left\{ -1.96 < \sqrt{n} \frac{(\bar{X} - \mu)}{\sigma} < 1.96 \right\} = .95$$

or, equivalently,

$$P \left\{ -1.96 \frac{\sigma}{\sqrt{n}} < \bar{X} - \mu < 1.96 \frac{\sigma}{\sqrt{n}} \right\} = .95$$

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$$\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}}$$

$$\left(\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \right)$$

is called a *95 percent confidence*

two-sided confidence interval.

One-sided CI

95 percent *one-sided upper confidence interval* for μ

$$P\{Z < 1.645\} = .95$$

As a result,

$$\left(\bar{x} - 1.645 \frac{\sigma}{\sqrt{n}}, \infty\right)$$

$$P\left\{\sqrt{n} \frac{(\bar{X} - \mu)}{\sigma} < 1.645\right\} = .95$$

lower confidence interval

or

$$P\left\{\bar{X} - 1.645 \frac{\sigma}{\sqrt{n}} < \mu\right\} = .95$$

$$\left(-\infty, \bar{x} + 1.645 \frac{\sigma}{\sqrt{n}}\right)$$

Example 7.3.a. Suppose that when a signal having value μ is transmitted from location A the value received at location B is normally distributed with mean μ and variance 4. That is, if μ is sent, then the value received is $\mu + N$ where N , representing noise, is normal with mean 0 and variance 4. To reduce error, suppose the same value is sent 9 times. If the successive values received are 5, 8.5, 12, 15, 7, 9, 7.5, 6.5, 10.5, let us construct a 95 percent confidence interval for μ .

Example 7.3.b. Determine the upper and lower 95 percent confidence interval estimates of μ in Example 7.3.a.

$$1.645 \frac{\sigma}{\sqrt{n}} = \frac{3.29}{3} = 1.097$$

the 95 percent upper confidence interval is

$$(9 - 1.097, \infty) = (7.903, \infty)$$

and the 95 percent lower confidence interval is

$$(-\infty, 9 + 1.097) = (-\infty, 10.097) \quad \blacksquare$$

Generalized CI Formula

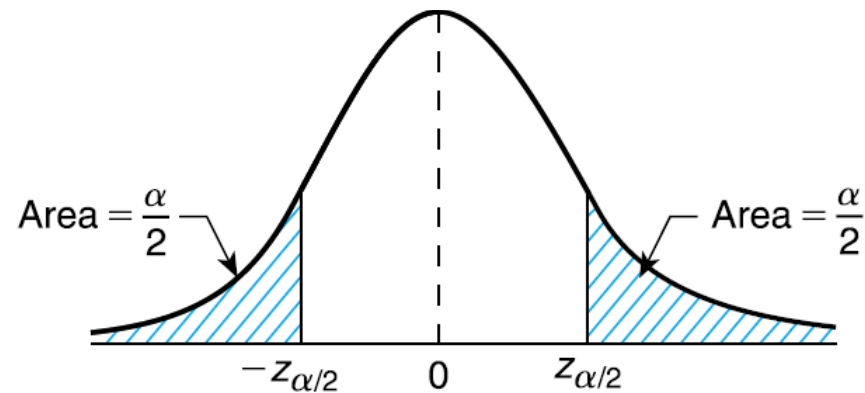
$$P\{-z_{\alpha/2} < Z < z_{\alpha/2}\} = 1 - \alpha$$

As a result, we see that

$$P\left\{-z_{\alpha/2} < \sqrt{n} \frac{(\bar{X} - \mu)}{\sigma} < z_{\alpha/2}\right\} = 1 - \alpha$$

or

$$P\left\{-z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \bar{X} - \mu < z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right\} = 1 - \alpha$$



$$P\{-z_{\alpha/2} < Z < z_{\alpha/2}\} = 1 - \alpha.$$

or

$$P\left\{-z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu - \bar{X} < z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right\} = 1 - \alpha$$

That is,

$$P\left\{\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right\} = 1 - \alpha$$

$$\left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \quad \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

$$\left(\bar{x} - z_{\alpha} \frac{\sigma}{\sqrt{n}}, \infty \right) \quad 100(1 - \alpha) \text{ percent one-sided upper}$$

and

$$\left(-\infty, \bar{x} + z_{\alpha} \frac{\sigma}{\sqrt{n}} \right) \quad 100(1 - \alpha) \text{ percent one-sided lower}$$

Example 7.3.c. Use the data of Example 7.3.a to obtain a 99 percent confidence interval estimate of μ , along with 99 percent one-sided upper and lower intervals.

Solution. Since $z_{.005} = 2.58$, and

$$2.58 \frac{\sigma}{\sqrt{n}} = \frac{5.16}{3} = 1.72$$

it follows that a 99 percent confidence interval for μ is

$$9 \pm 1.72 \qquad (7.28, 10.72).$$

since $z_{.01} = 2.33$

99 percent upper confidence interval is

$$(9 - 2.33(2/3), \infty) = (7.447, \infty)$$

99 percent lower confidence interval is

$$(-\infty, 9 + 2.33(2/3)) = (-\infty, 10.553)$$

CI with a certain level

compute an interval of length .1 with 99 percent confidence,

99 percent confidence interval for μ

$$\left(\bar{x} - 2.58 \frac{\sigma}{\sqrt{n}}, \quad \bar{x} + 2.58 \frac{\sigma}{\sqrt{n}} \right) \quad (7.28, 10.72)$$

Thus, to make the length to .1,

$$5.16 \frac{\sigma}{\sqrt{n}} = .1$$

or

$$n = (51.6 \sigma)^2$$

Hence, its length is

$$5.16 \frac{\sigma}{\sqrt{n}}$$

Example 7.3.d. From past experience it is known that the weights of salmon grown at a commercial hatchery are normal with a mean that varies from season to season but with a standard deviation that remains fixed at 0.3 pound. If we want to be 95 percent certain that our estimate of the present season's mean weight of a salmon is correct to within ± 0.1 pound, how large a sample is needed?

A 95 percent confidence interval estimate

$$\mu \in \left(\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \right)$$

$\bar{x}$ is within $1.96(\sigma/\sqrt{n}) = .588/\sqrt{n}$

95 percent certain that $\bar{x}$ is within 0.1 of μ

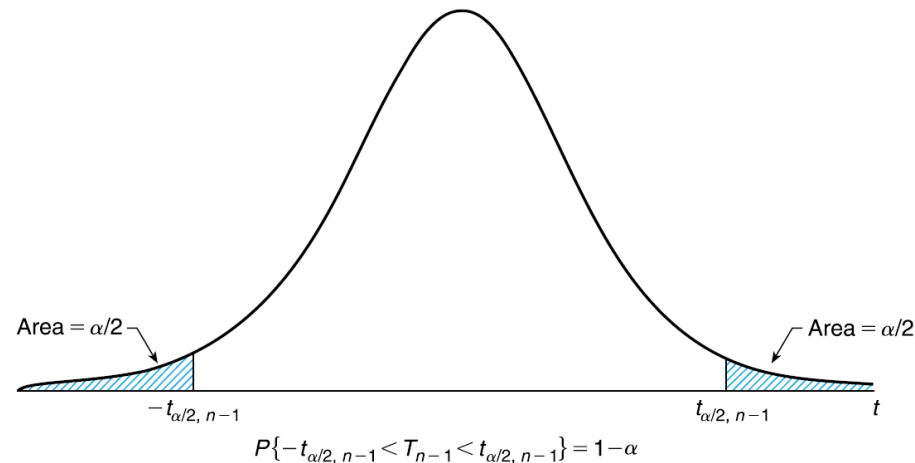
$$\frac{.588}{\sqrt{n}} \leq 0.1$$

That is, provided that

$$n \geq 34.57$$

$$\sqrt{n} \geq 5.88$$

CI for unknown Variance



$\sqrt{n}(\bar{X} - \mu)/\sigma$ is a standard normal

$$S^2 = \sum_{i=1}^n (X_i - \bar{X})^2 / (n - 1)$$

$\sqrt{n} \frac{(\bar{X} - \mu)}{S}$ t -random variable with $n - 1$ degrees of freedom

$$P \left\{ -t_{\alpha/2, n-1} < \sqrt{n} \frac{(\bar{X} - \mu)}{S} < t_{\alpha/2, n-1} \right\} = 1 - \alpha$$

or, equivalently,

$$P \left\{ -t_{\alpha/2, n-1} \frac{S}{\sqrt{n}} < \bar{X} - \mu < t_{\alpha/2, n-1} \frac{S}{\sqrt{n}} \right\} = 1 - \alpha$$

$$P \left\{ \bar{X} - t_{\alpha/2, n-1} \frac{S}{\sqrt{n}} < \mu < \bar{X} + t_{\alpha/2, n-1} \frac{S}{\sqrt{n}} \right\} = 1 - \alpha$$

$$\mu \in \left(\bar{x} - t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}, \bar{x} + t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} \right)$$

Example 7.3.e. Let us again consider Example 7.3.a but let us now suppose that when the value μ is transmitted at location A then the value received at location B is normal with mean μ and variance σ^2 but with σ^2 being unknown. If 9 successive values are, as in Example 7.3.a, 5, 8.5, 12, 15, 7, 9, 7.5, 6.5, 10.5, compute a 95 percent confidence interval for μ .

$$\bar{x} = 9$$

and

$$s^2 = \frac{\sum x_i^2 - 9(\bar{x})^2}{8} = 9.5 \qquad s = 3.082$$

as $t_{.025,8} = 2.306$, a 95 percent confidence interval for μ

$$\left[9 - 2.306 \frac{(3.082)}{3}, 9 + 2.306 \frac{(3.082)}{3} \right] = (6.63, 11.37)$$

Upper and Lower CI

$$P \left\{ \sqrt{n} \frac{(\bar{X} - \mu)}{S} < t_{\alpha, n-1} \right\} = 1 - \alpha$$

or

$$P \left\{ \bar{X} - \mu < \frac{S}{\sqrt{n}} t_{\alpha, n-1} \right\} = 1 - \alpha$$

or

$$P \left\{ \mu > \bar{X} - \frac{S}{\sqrt{n}} t_{\alpha, n-1} \right\} = 1 - \alpha$$

$$\mu \in \left(\bar{x} - \frac{s}{\sqrt{n}} t_{\alpha, n-1}, \infty \right)$$

$$\mu \in \left(-\infty, \bar{x} + \frac{s}{\sqrt{n}} t_{\alpha, n-1} \right)$$

Example 7.3.f. Determine a 95 percent confidence interval for the average resting pulse of the members of a health club if a random selection of 15 members of the club yielded the data 54, 63, 58, 72, 49, 92, 70, 73, 69, 104, 48, 66, 80, 64, 77. Also determine a 95 percent lower confidence interval for this mean.

```
>d = c(54, 63, 58, 72, 49, 92, 70, 73, 69, 104, 48, 66, 80, 64, 77)
>l = mean(d) - qt(.975, 14) * sqrt(var(d)/15)
>u = mean(d) + qt(.975, 14) * sqrt(var(d)/15)
>c(l, u)
[1] 60.8669366732691, 77.66639666600643
```

CI for Variance of Normal Population

If $X_1, \dots, X_n$ is a sample from a normal distribution μ and σ^2

we can construct a confidence interval for σ^2

$$(n-1) \frac{S^2}{\sigma^2} \sim \chi_{n-1}^2$$

$$P \left\{ \chi_{1-\alpha/2, n-1}^2 \leq (n-1) \frac{S^2}{\sigma^2} \leq \chi_{\alpha/2, n-1}^2 \right\} = 1 - \alpha$$

$$P \left\{ \frac{(n-1)S^2}{\chi_{\alpha/2, n-1}^2} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi_{1-\alpha/2, n-1}^2} \right\} = 1 - \alpha$$

when $S^2 = s^2$, a $100(1 - \alpha)$ percent confidence interval for σ^2

$$\left(\frac{(n-1)s^2}{\chi_{\alpha/2, n-1}^2}, \frac{(n-1)s^2}{\chi_{1-\alpha/2, n-1}^2} \right)$$

Example 7.3.i. A standardized procedure is expected to produce washers with very small deviation in their thicknesses. Suppose that 10 such washers were chosen and measured. If the thicknesses of these washers were, in inches,

.123 .133

.124 .125

.126 .128

.120 .124

.130 .126

90 percent confidence interval for the standard deviation

```
>d = c(.123, .124, .126, .120, .130, .133, .125, .128, .124, .126)
```

```
>l = 9 * var(d)/ qchisq(.95, 9)
```

```
>u = 9 * var(d)/ qchisq(.05, 9)
```

```
>c(l, u)
```

```
[1]7.26403231164731e-06 3.69611516361794e-05
```

It follows that, with confidence .90,

$$\sigma^2 \in (7.2640 \times 10^{-6}, 36.9612 \times 10^{-6})$$

Taking square roots yields, with confidence .90,

$$\sigma \in (.00269, .00608) \quad \blacksquare$$

CI for the Differences in Means of 2 Normal Populations

$X_1, X_2, \dots, X_n$ be a sample of size n from a normal population
mean μ_1 and variance σ_1^2 ;

$Y_1, \dots, Y_m$ be a sample of size m
mean μ_2 and variance σ_2^2

interested in estimating $\mu_1 - \mu_2$.

$$\bar{X} = \sum_{i=1}^n X_i/n \text{ and } \bar{Y} = \sum_{i=1}^m Y_i/m$$

$\bar{X} - \bar{Y}$ maximum likelihood estimator of $\mu_1 - \mu_2$

$$\bar{X} \sim \mathcal{N}(\mu_1, \sigma_1^2/n)$$

$$\bar{Y} \sim \mathcal{N}(\mu_2, \sigma_2^2/m)$$

sum of independent normal random variables

$$\bar{X} - \bar{Y} \sim \mathcal{N}\left(\underbrace{\mu_1 - \mu_2}, \underbrace{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}\right)$$

assuming σ_1^2 and σ_2^2 are known

$$\frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}} \sim \mathcal{N}(0, 1)$$

$$P \left\{ -z_{\alpha/2} < \frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}} < z_{\alpha/2} \right\} = 1 - \alpha$$

$$P \left\{ \bar{X} - \bar{Y} - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}} < \mu_1 - \mu_2 < \bar{X} - \bar{Y} + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}} \right\} = 1 - \alpha$$

two-sided confidence interval estimate for $\mu_1 - \mu_2$ is

$$\mu_1 - \mu_2 \in \left(\bar{x} - \bar{y} - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}, \bar{x} - \bar{y} + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}} \right)$$

100(1 - α) percent one-sided

$$\mu_1 - \mu_2 \in \left(-\infty, \bar{x} - \bar{y} + z_{\alpha} \sqrt{\sigma_1^2/n + \sigma_2^2/m} \right)$$

Example 7.4.a. Two different types of electrical cable insulation have recently been tested to determine the voltage level at which failures tend to occur. When specimens were subjected to an increasing voltage stress in a laboratory experiment, failures for the two types of cable insulation occurred at the following voltages:

<i>Type A</i>		<i>Type B</i>	
36	54	52	60
44	52	64	44
41	37	38	48
53	51	68	46
38	44	66	70
36	35	52	62
34	44		

```
>l = mean(x) - mean(y) - qnorm(.975) * sqrt(40/14 + 100/12)
>u = mean(x) - mean(y) + qnorm(.975) * sqrt(40/14 + 100/12)
>c(l, u)
[1]6.491114 19.604124
```

population variances σ_1^2 and σ_2^2 are unknown

$$S_1^2 = \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{n-1}$$

$$S_2^2 = \sum_{i=1}^m \frac{(Y_i - \bar{Y})^2}{m-1}$$

Table 7.1 100(1 - α) Percent Confidence Intervals

$$X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2) \quad \bar{X} = \sum_{i=1}^n X_i / n, \quad S = \sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 / (n-1)}$$

Assump- tion	Parame- ter	Confidence Interval	Lower Interval	Upper Interval
σ^2 known	μ	$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$	$\left(-\infty, \bar{X} + z_{\alpha} \frac{\sigma}{\sqrt{n}}\right)$	$\left(\bar{X} + z_{\alpha} \frac{\sigma}{\sqrt{n}}, \infty\right)$
σ^2 unknown	μ	$\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}$	$\left(-\infty, \bar{X} + t_{\alpha, n-1} \frac{S}{\sqrt{n}}\right)$	$\left(\bar{X} - t_{\alpha, n-1} \frac{S}{\sqrt{n}}, \infty\right)$
μ unknown	σ^2	$\left(\frac{(n-1)S^2}{\chi_{\alpha/2, n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\alpha/2, n-1}^2}\right)$	$\left(0, \frac{(n-1)S^2}{\chi_{1-\alpha, n-1}^2}\right)$	$\left(\frac{(n-1)S^2}{\chi_{\alpha, n-1}^2}, \infty\right)$